Abdelrahman Zahran

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Professional Summary

Computer Engineering student with extensive experience in UI/UX development, ReactJS, and Tailwind CSS, delivering scalable, user-friendly applications. Proficient in building modular, responsive interfaces and integrating APIs for seamless functionality. Skilled in machine learning and NLP, with a focus on explainable AI for legal applications. Having a proven track record in delivering feature-based solutions using Agile methodologies.

Education

Computer Engineering (English) | Marmara University, Istanbul, Türkiye | 10/2021 - Present

- Thesis: *Legally Speaking: Developing Explainable AI Approaches for Legal Decision-Making*, focusing on NLP and transparent AI for legal text analysis.
- Relevant Coursework: Java Programming, C Programming, Data Structures, Object Oriented Programming (OOP), Database Systems, Software Engineering, Machine Learning, Artificial Intelligence, Data Science, Natural Language Processing (NLP)

Work Experience

Application Software Developer

EFA Bilişim Yazılım | Istanbul, Türkiye | 07/2024 - 10/2024

- Developed an AI-driven Task Management system using Next.js, React.js, and Tailwind CSS, improving operational efficiency for a client organization.
- Engineered responsive UI components with Shadcn UI and Tailwind CSS, ensuring a seamless user experience across desktop and mobile platforms.
- Integrated APIs using Next.js Server Actions and Convex for optimized data handling and real-time updates.
- Implemented secure authentication with Next Auth, enhancing user management and access control.
- Delivered modular, reusable code for task management, project organization, and sidebar navigation, adhering to software design best practices.

Skills: TypeScript, Next.js, React.js, Tailwind CSS, Shadcn UI, Convex, Next Auth, API Integration, Modular Design.

Machine Learning Intern

FLYtelligence Software Solutions (Marmara Teknopark) | Istanbul, Türkiye | 07/2023 - 10/2023

- Built an AI-integrated barcode detection system using React.js for the frontend, improving accuracy by 30% and reducing processing time by 40%.
- Developed user-facing web applications with React.js and Node.js, enhancing data management for 500+ users.
- Conducted research on machine learning techniques, integrating findings into UI/UX improvements for better usability.

Skills: React.js, Node.js, Python, PyTorch, TensorFlow, UI/UX Design, API Integration.

Key Projects

Machine Learning in Law

- Analyzed 14,849 Turkish Constitutional Court decisions to predict "violation" and "no violation" outcomes.
- Data Preprocessing and EDA: Cleaned and normalized text data, visualized decision types, and explored data patterns.
- Model Development: Implemented baseline models (SVM, Decision Tree, Logistic Regression) and advanced deep learning models (LSTM, CNN, Transformers) for text classification tasks.
- Model Evaluation and Optimization: Assessed models using metrics like accuracy, F1-score, and precision-recall, applying techniques such as cross-validation and grid search for hyperparameter tuning.
- Explainable AI (XAI): Leveraged SHAP, LIME, and EBM to enhance model interpretability, identifying key decision factors for legal professionals.

Skills: Python Programming (TensorFlow, PyTorch, Scikit-learn), Machine Learning and Deep Learning, Natural Language Processing (NLP), Data Preprocessing and Analysis, Explainable AI (SHAP, LIME, EBM), Model Evaluation and Optimization, Legal Text Analysis.
Link: github.com/Pirate-960/Machine-Learning-in-Law

Reversi (Othello) Game

- Multi-Version AI Development: Designed four AI versions for Othello, implementing heuristic-based strategies (disc count, mobility, stability) and advanced pattern recognition for strategic depth.
- Search Algorithm Optimization: Developed a Negamax-based engine with Alpha-Beta pruning, transposition tables, killer moves, and iterative deepening for enhanced decision-making under time constraints.
- Game Engine & Analytics: Built a modular Othello engine with real-time logging and JSON output for detailed post-game analysis.
- Performance Evaluation: Conducted extensive benchmarking to assess heuristic effectiveness and optimize computational efficiency.

Skills: Python, Heuristic Development, Algorithm Optimization, Search Algorithms, Iterative Deepening, Object-Oriented Programming, JSON Serialization, System Architecture, Performance Testing, Benchmarking.
Link: github.com/Pirate-960/Reversi-Game

VibeCheck: Video Sentiment Analysis System

- Developed a SaaS platform for multimodal video sentiment analysis using Next.js, React.js, and Tailwind CSS, enabling user-friendly video uploads and result visualization.
- Designed functional requirements and class diagrams for a responsive web dashboard, integrating APIs for video processing and sentiment analysis.
- Implemented user authentication with Auth.js and API quota management, ensuring secure and scalable access.
- Contributed to non-functional requirements, achieving 85% model accuracy and sub-5-second processing for 30-second videos.

Skills: Next.js, React.js, Tailwind CSS, Auth.js, Prisma, API Integration, UI/UX Design
Links:

- Frontend Hub: github.com/Vibe-Check-Ai/VibeCheck-Hub
- AI Models: github.com/Vibe-Check-Ai/VibeCheck-DeepFeel
- Documentation: github.com/Vibe-Check-Ai/VibeCheck-Docs

Technical Skills

- **Frontend Development**: React.js, Next.js, TypeScript, JavaScript, Tailwind CSS, Shadcn UI, HTML, CSS.
- **Backend & Database**: Node.js, Convex, MongoDB, SQLite, Microsoft SQL Server, Prisma, API Integration.
- **Machine Learning & AI**: Python, PyTorch, TensorFlow, Scikit-Learn, NLP, Explainable AI (SHAP, LIME).
- **Tools**: Git, GitHub, Jira, JSON, Jupyter Notebook.

Certifications

- Machine Learning Specialization, Stanford & DeepLearning.AI
- Deep Learning Specialization, DeepLearning.AI
- Natural Language Processing Specialization, DeepLearning.AI

Languages

- Arabic (Native), English (C1), Turkish (B1), German (B2)